Ziyad Fouda *AI Engineer*

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OBJECTIVE

Motivated and detail-oriented AI Engineering student seeking opportunities to apply expertise in machine learning, deep learning, and intelligent system development to real-world challenges. Passionate about designing, building, and deploying AI-powered solutions, with a strong interest in generative AI, large language models, and scalable AI systems that drive innovation and solve complex problems.

EDUCATION

Computer and Control Systems Engineering — 2021 – 2026
Mansoura University — Mansoura, Egypt

PROFESSIONAL EXPERIENCE

AI & Large Language Models Intern — 12/2025 – 02/2026
Tips Hindawi — remote

AI & Data Science - Microsoft Machine Learning Engineer Intern — 06/2025 – 12/2025
Digital Egypt Pioneers Program — remote

AI Internship Trainee — 01/2025 – 02/2025
National Telecommunication Institute — Mansoura, Egypt

CERTIFICATES

- Machine Learning Specialization - deeplearning.ai
- AI Engineering Specialization - Scrimba
- Python Data Association - DataCamp

SKILLS

Languages & Tools
python, JavaScript, SQL, Docker

Soft Skills
Creativity, Problem-Solving, Time Management

Frameworks/Libraries
scikit-learn, pyTorch, Langchain, OpenAI SDK, LangGraph

PROJECTS

GenLearn
AI-powered learning platform that converts user documents into structured course materials with sections and lecture videos
• Developed an AI-based educational platform that transforms uploaded documents into organized course content, including lessons, sections, and video-based lectures.
• Built lecture content generation workflows using modern AI technologies such as **LangChain** and **Azure AI Foundry** to create coherent and engaging learning material.
• Contributed to improving text-to-speech (TTS) audio output using **Azure services** to make the generated lectures sound more natural and clear.
• Worked on integrating AI-driven content generation into a full application workflow aimed at making educational resources more accessible and scalable.

PopChoice
RAG-based recommendation application using embeddings and Large Language Models
- Developed an AI-powered movie recommendation system that suggests personalized movies based on users' preferences and motivations.
- Implemented **Retrieval-Augmented Generation (RAG)** by embedding user inputs and matching them against a vector database of movie embeddings stored in **Supabase**.
- Built and deployed the backend using **Cloudflare Workers** to provide scalable, low-latency recommendations.
- Designed and hosted a responsive frontend using **HTML, CSS, JavaScript,** and **GitHub Pages.**

Research Agent with LangGraph 🔗
Agentic Research Workflow | LangGraph | LangChain | Azure OpenAI | Tavily
- Developed an agentic research workflow using **LangGraph** that decomposes user topics into research plans, generates research drafts, and evaluates the results through multiple LLM-powered stages.
- Implemented a Planner **->** Writer **->** Reviewer workflow with conditional routing, allowing the reviewer to send drafts back to the writer for iterative improvement.
- Integrated **Tavily** as a web search tool for the research writer and **Azure OpenAI** for planning, content generation, and structured review.
- Designed a typed shared state to manage research plans, drafts, review feedback, retry attempts, and workflow control across LangGraph nodes.

Study Tracker App 🔗
Full-Stack Web Application | FastAPI | PostgreSQL | Docker
- Developed a full-stack study tracking application for recording study sessions, organizing users, and analyzing study-time statistics.
- Built a RESTful backend with **FastAPI**, **Pydantic**, and **psycopg2**, including user management, study-session tracking, date-range queries, and statistics endpoints.
- Designed and integrated a **PostgreSQL** database with relational constraints, initialization scripts, sample data, and persistent Docker volumes.
- Containerized the Frontend, Backend, and PostgreSQL services using **Docker Compose** with service health checks and inter-service networking.

Liver Cirrhosis Stage Prediction 🔗
Machine Learning Healthcare Application | XGBoost | Streamlit
- Developed a machine learning model to predict liver cirrhosis severity with **97% accuracy**.
- Applied data preprocessing techniques, including outlier removal and feature scaling.
- Evaluated multiple algorithms and selected **XGBoost** for optimal performance.
- Deployed the solution as an interactive **Streamlit** application for real-time predictions.

Real-Time Sign Language Detection System 🔗
Mediapipe | Random Forest | Streamlit
- Developed a real-time sign language recognition system using **Mediapipe** hand-tracking and a **Random Forest** classifier.
- Created a custom dataset of hand landmarks (~300 samples per gesture) to improve model accuracy and reliability.
- Extracted and processed **21 hand landmarks (42 features)** for gesture classification.
- Built an interactive **Streamlit** application for live camera input and real-time gesture prediction.